

An Automatic Following/Forming Framework for Inverter-Based Resources

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Why a Converter Must Sometimes Lead

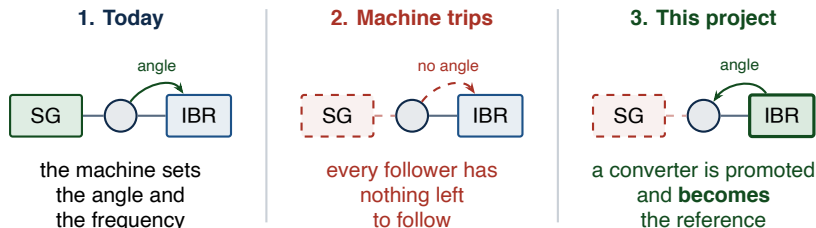


Figure 1: A synchronous generator (SG) sets the voltage angle; an inverter-based resource (IBR) normally follows it.

- 1** One in-house chain: power flow (PF) → time-domain simulation (TS).
- 2** Four converters, one plant, two switchable controllers.
- 3** **How many** must lead is decided by the network, not fixed by hand.
- 4** Verified through trip, fault, and reconnection.

Power Flow and Time-Domain Formulation

PF — one steady operating point

$$x = \begin{bmatrix} \theta \\ |V| \end{bmatrix} \quad (1)$$

$$g(x) = \begin{bmatrix} P^{\text{inj}} - P(x) \\ Q^{\text{inj}} - Q(x) \end{bmatrix} = \mathbf{0} \quad (2)$$

$$x^{(r+1)} = x^{(r)} - J^{-1} g(x^{(r)}) \quad (3)$$

$$J = \partial g / \partial x \quad (4)$$

No time appears. This point supplies the healthy reference V_i^* per bus and the warm start for the dynamic model.

TS — the same network, stepped in time

$$\dot{x} = f(x, y, t), \quad \mathbf{0} = g(x, y, t) \quad (5)$$

$$X = \begin{bmatrix} x \\ y \end{bmatrix}, \quad g = Y_{\text{net}}(t) y - I_{\text{bus}}(x, y) \quad (6)$$

Implicit trapezoidal step, $h = t_{n+1} - t_n$:

$$x_{n+1} = x_n + \frac{h}{2} [f_n + f_{n+1}] \quad (7)$$

$$\mathbf{0} = g(x_{n+1}, y_{n+1}, t_{n+1}) \quad (8)$$

$f_n = f(x_n, y_n, t_n)$; (7) and (8) are solved together for X_{n+1} by damped Newton to $\|r\|_{\infty} < 10^{-8}$. Time enters through the events: $Y_{\text{net}}(t)$ is re-stamped at each one, and $m_i(t)$ picks each converter's controller branch.

One Converter, Two Controller Branches

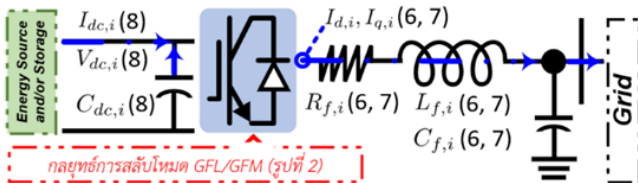


Figure 2: One IBR: a shared power stage — direct-current (DC) link, switching bridge, alternating-current (AC) filter — with the GFL/GFM controller branch selected at run time.

$$\mathbf{X}_{sys} = [\mathbf{x}_{IBR,1} \ \mathbf{x}_{IBR,2} \ \mathbf{x}_{IBR,3} \ \mathbf{x}_{IBR,4} \ \mathbf{x}_{SG}]^T \quad (9)$$

$$\mathbf{x}_{IBR} = \underbrace{\begin{bmatrix} i_d, i_q, V_{dc} \end{bmatrix}}_{\text{shared plant}} \underbrace{\begin{bmatrix} \mathbf{x}_{GFL} \ \mathbf{x}_{GFM} \end{bmatrix}}_{\text{source}} \underbrace{\begin{bmatrix} I_{dc} \end{bmatrix}}_{\text{source}}, \quad \mathbf{x}_{SG} = [\delta, \omega, E'_q, E'_d, E''_q, E''_d] \quad (10)$$

$$\mathbf{x}_{GFL} = [\theta_{PLL}, \xi_{PLL}, \xi_P, \xi_Q, \xi_{Id}, \xi_{Iq}], \quad \mathbf{x}_{GFM} = [\theta_{VSG}, \omega_{VSG}, E, \xi_{Vd}, \xi_{Vq}, \xi_{Id}, \xi_{Iq}] \quad (11)$$

How the Mode Switches

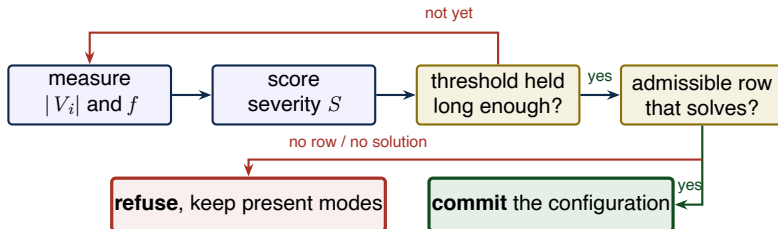


Figure 3: The decision path: a threshold crossing is a request, and the last box can refuse it.

What is measured and scored

$$J_V = \frac{||V_i| - V_i^*|}{\Delta V_{base}} \quad (12)$$

$$J_f = \frac{|f_{coi} - f_0|}{\Delta f_{base}} \quad (13)$$

$$S = \text{sat}_{[0,1]} \left(\frac{1}{2} J_V + \frac{1}{2} J_f \right) \quad (14)$$

What the command is, in this framework

$$m_i(t) \in \{\text{follow, lead, out}\} \quad (15)$$

$$\dot{x}_i = f_{GFL}, \quad f_{GFM}, \quad \text{or } 0 \quad (16)$$

One branch runs, the other is held — and the converters are **not** switched independently: the whole configuration comes from a table screened in advance, or is refused.

Lead at $S \geq 0.65$ held 0.10 s; follow at $S \leq 0.35$

Network and Case Study: IEEE 14-Bus

Resources	
Unit	Bus
SG_1	1
IBR_1	2
IBR_2	3
IBR_3	6
IBR_4	8

SG_1 is the slack bus and the only synchronous reference. IBR_1 – IBR_4 are dual-mode converters.

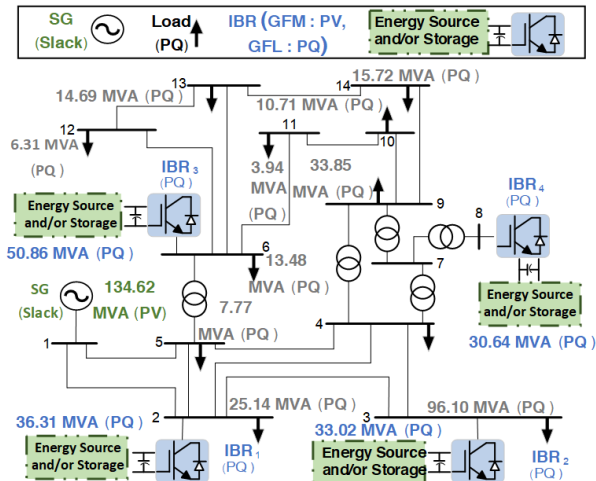


Figure 4: One SG at bus 1 and four dual-mode IBRs at buses 2, 3, 6 and 8.